

GETTING STARTED WITH PYTHON

Building a simple calculator Gui with Tkinter

Technology Chosen

Python with Tkinter

Why I Chose It

1. Easy for beginners and something I have had an interest in learning
2. Built into Python (no extra installation needed)
3. Helps understand GUI development concepts for much cleaner and user-friendly system

End Goal:

To build a simple calculator with a graphical user interface (GUI) that can perform basic arithmetic operations and display results interactively

Quick Summary of the Technology

What is it?

Tkinter is a standard Python library used to create graphical user interfaces (GUIs)

Where is it used?

1. Desktop applications
2. Educational tools
3. Simple system utilities

Real-world Example:

Tkinter being a standard library that comes pre-installed on Python is used by Developers to build a wide range of custom applications, especially internal tools and prototypes where functionality is more important than highly custom aesthetics e.g., QR Code Generators and Detector and Age Calculators

System Requirements

- Operating System: Windows python
- Editor: Visual Studio Code
- Language: Python (version 3.14.3)
- Packages: Tkinter which is pre-installed on python
- AI: Chatgpt and Copilot

Installation & Setup Instructions

1. Install Python
2. Confirm installation – Type *python --version* on PowerShell
3. Download vscode
4. Install python extension on vscode

Minimal Working Example

This program creates a simple calculator GUI where users input an expression and get a result when they click a button.

Code with inline comments

```
import tkinter as tk

def calculate ():
    try:
        result = eval(entry.get())
        output.config(text="Result: " + str(result))
    except:
        output.config(text="Error", fg="red")

# Create window
app = tk.Tk()
app.title("Calculator 🧮")
app.geometry("300x200")
app.configure(bg="#FFC0CB") # light pink background

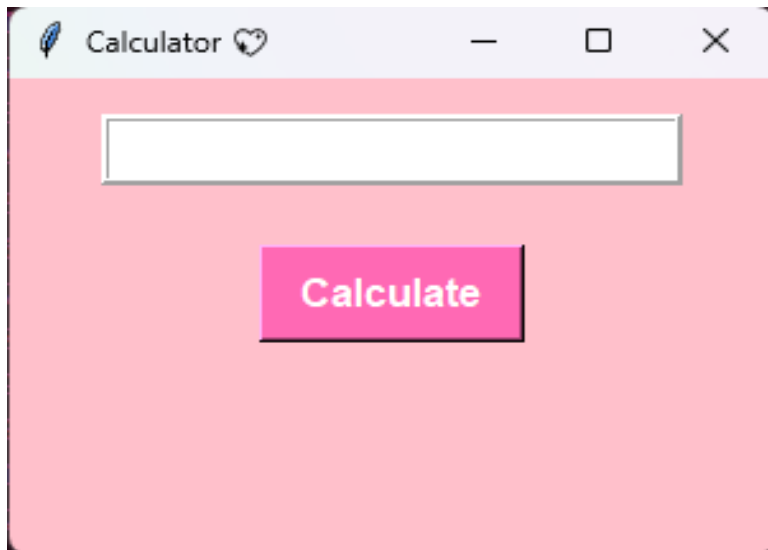
# Input field
entry = tk.Entry(app, font=("Arial", 14), bd=3, relief="ridge", justify="center")
entry.pack(pady=15)

# Add Calculate Button
btn = tk.Button(
    app,
    text="Calculate",
    command=calculate,
    bg="#FF69B4", # hot pink
    fg="white",
    font=("Arial", 12, "bold"),
    padx=10,
    pady=5
)
btn.pack(pady=10)

# Output label
output = tk.Label(
    app,
    text="",
    bg="#FFC0CB",
```

```
fg="black",  
font=("Arial", 12)  
)  
output.pack(pady=10)  
  
# Run app  
app.mainloop()
```

Expected Output



AI PROMPTS

Prompt 1

Give me a Beginner friendly guide to learning Python

AI response with Topic break down and guide on how to setup the environment eg Python Installation

Prompt 2

Help me understand TKINTER and how to incorporate it

AI Response with clear steps on importing library, basic structure and building block of Tkinter

Prompt 3

How can a string be executed as Python code and return the result. Give clear example of functions that can be used and how

Response with definitions and examples on:

- `eval()`
- `ast.literal_eval()`
- Safer libraries like `numexpr`
- Parse the mathematical expression manually.

AI Helpfulness

Learning with AI was much easier and helpful cause it gave clear instructions and additional materials to read from

Common Issues & Fixes

Issue 1

Had issues running the python code

Solution: Noticed I had not installed python on my machine

Issue 2

Using `eval()` is risky since it can execute unsafe code

Solution: Use safer parsing methods e.g.: `ast.literal_eval()`

Issues 3

The calculator ui/ux was off and not appealing

Solution: Asked chatgpt how to change color and display dimensions for the calculator output

References

Foundation, Python Software. 2001. "Python interface to Tcl/Tk."
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